INCH-POUND
MIL-DTL-24643/36F
1 October 2009
SUPERSEDING
MIL-DTL-24643/36E
22 August 2002

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, -20 °C TO +90 °C, TYPE LS3SWU (INCLUDING VARIATION LS3SWUS)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24643.

This cable can be manufactured in two variations: LS3SWU and LS3SWUS (double overall shield).

Construction, Watertight

First	-	Copper conductor, coated or uncoated, 18 AWG size, Class B stranding (7 by 0.0152 inches).
Second	-	Separator may be used at manufacturer's option where required to provide free-stripping insulation.
Third	-	Thermoset insulation. Nominal wall thickness of 0.023 inch; wall thickness may vary from nominal as necessary, in order for the completed cable to meet the specified electrical requirements. Colored insulation, one white, one black, and one red conductor for each triad.
Fourth	-	Three conductors (one black, one white, and one red) cabled together with a nominal lay of 3.5 inches to form a triad.
Fifth	-	An optional binder.
Sixth	-	Braided shield of 34 or 36 AWG, tin-coated copper. Braid angle of 30 to 35 degrees; minimum coverage of 85 percent.
Seventh	-	Shield insulation of two polyester tapes, outer tape sealed. An alternate shield insulation may consist of one polyester tape, plus an extruded jacket of a suitable transparent material, minimum average thickness of the transparent material 0.003 inch. The standard identification code shall be applied by Method 2.
Eighth	-	The specified number of triads (see table I) shall be cabled together in accordance with 3.4.4 of MIL-DTL-24643. Cabling sequence shall be consecutive starting with no. 1 from the center outward. Fillers may be used as necessary to form a firm well-rounded assembly.
Ninth	-	An optional binder.

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Construction, Variant for Type LS3SWU

Tenth - Cross-linked polyolefin jacket (see table I for thickness).

Construction, Variant for Type LS3SWUS

Tenth - Braided shield of tin-coated copper with a minimum coverage of 85 percent.

Eleventh - Spirally-wrapped 0.002-inch thick polyethylene terephthalate (polyester film) separator tape with 25 percent minimum overlap.

Twelfth - Braided shield of tin-coated copper with a minimum coverage of 85 percent.

Thirteenth - An optional binder.

Fourteenth - Cross-linked polyolefin jacket (see table I for thickness).

TABLE I. Details.

Military part no. M24643/36	Type and size	Number of triads	Cable jacket thickness min. avg. (inch)	Overall diameter	
				min. (inch)	max. (inch)
-01UN	LS3SWU-3	3	0.040	0.610	0.655
-01UD	LS3SWUS-3	3	0.040	0.675	0.710
-02UN	LS3SWU-7	7	0.040	0.880	0.940
-02UD	LS3SWUS-7	7	0.040	0.940	1.000
-03UN	LS3SWU-10	10	0.050	1.100	1.180
-03UD	LS3SWUS-10	10	0.050	1.160	1.240
-04UN	LS3SWU-14	14	0.050	1.200	1.280
-04UD	LS3SWUS-14	14	0.050	1.260	1.340
-05UN	LS3SWU-19	19	0.050	1.370	1.450
-05UD	LS3SWUS-19	19	0.050	1.430	1.510
-06UN	LS3SWU-24	24	0.050	1.640	1.760
-06UD	LS3SWUS-24	24	0.050	1.700	1.820
-07UN	LS3SWU-30	30	0.050	1.760	1.860
-07UD	LS3SWUS-30	30	0.050	1.820	1.920
-08UN	LS3SWU-37	37	0.060	1.890	1.940
-08UD	LS3SWUS-37	37	0.060	1.950	2.050
-09UN	LS3SWU-44	44	0.060	2.140	2.240
-09UD	LS3SWUS-44	44	0.060	2.200	2.300

REQUIREMENTS:

Qualification required.

INSPECTION:

Basic Electricals:

Conductor resistance (ohms/1000 feet at 25 °C, max.) 7.06

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Voltage withstand (volts, root mean square, min.)	
Conductor to conductor	2000
Conductor to shield	1000
Component shield to component shield	500
Overall shield to overall shield (Type LS3SWUS only)	200
Insulation resistance (megohms/1000 feet, min.)	
Conductor to conductor	500
Conductor to shield	500
Component shield to component shield	100
Overall shield to overall shield (Type LS3SWUS only)	100
Conductor continuity	No failure
Shield continuity	No failure
Jacket flaws (Type LS3SWUS only)	No failure
Group A:	
Visual and dimensional	No failure
Hydrostatic (open end) (leakage at 300 lb/in^2, 2 hours, in^3, max.)	1.0
Capacitance	
Mutual capacitance, 1 kilohertz (KHz) (pF/ft, max.)	30
Capacitance unbalanced (percent max.)	10
Group B:	
Thermoset proof test (percent, max.)	
Insulation	50
Jacket (when tested at 200 °C)	50
Drip (95±1 °C)	Zero
Physicals (unaged)	
Insulation (extruded)	
Tensile strength (lb/in^2, min.)	700
Elongation (percent, min.)	150
Jacket (cable)	
Tensile strength (lb/in^2, min.)	1300
Elongation (percent, min.)	160
Tear (lb/in thickness, min.)	35

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Group C:	
Physicals (aged) air oven	
Insulation (extruded)	
Tensile strength (percent of unaged, min.)	75
Elongation (percent of unaged, min.)	75
Jacket (cable)	
Tensile strength (percent of unaged, min.)	60
Elongation (percent of unaged, min.)	60
Permanence of printing (jacket) (cycles, min.)	125
Heat distortion (percent of unaged, max.)	30
Cable sealant removability	No failure
Shrinkage	No failure
Shield (conformance to material, construction and coverage)	No failure
Surface transfer impedance (Type LS3SWUS only)	
Milliohms per meter (max.)	700
EMP response (dB, min.)	60
Group D:	
Flame propagation (cable)	No failure
Qualification Inspection:	
Qualification inspection shall include basic electricals; groups A, B, C, and D; plus the following:	
Aging and compatibility (cable) (125±5 °C)	No failure
Abrasion resistance (jacket) (scrapes, min.)	75
Acid gas equivalent (percent, max.)	
Jacket	2
Fillers	2
Insulation	18
Shield insulation	2
Halogen content (percent, max.)	
Jacket	0.2
Fillers	0.2
Insulation	0.2
Immersion (jacket)	
Tensile strength (percent of unaged, min.)	50

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Elongation (percent of unaged, min.)	50
Smoke index (max.)	
Jacket	25
Fillers	45
Insulation	45
Toxicity index (max.)	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - Type A (hardness, min.)	80
Weathering (jacket)	No failure
Electrical moisture absorption	No failure

UNIT ORDERING LENGTHS:

Types	Sizes	Feet (nominal)
LS3SWU and LS3SWUS	3 through 19	1500
LS3SWU and LS3SWUS	24 through 44	500

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army – MI
Navy – SH

Preparing Activity:
Navy – SH
(Project 6145-2008-040)

Review Activities:
Army – AR, AV, CR
Navy – CG, EC
DLA – CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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